**W. Ryan James**
Research Assistant Professor
Florida International University
**Curriculum Vitae, August 10, 2026**
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## EDUCATION

**2020** — **PhD Environmental and Evolutionary Biology**
University of Louisiana at Lafayette (Advisor: James A. Nelson)

**2016** — **MS Biology**
University of Alabama at Birmingham (Advisor: James B. McClintock)

**2013** — **BS Biology**
University of Alabama (Advisor: Ryan L. Earley)

## EMPLOYMENT

**2025-Current** — **Research Assistant Professor -** Institute of Environment
Florida International University, *Miami, FL*

**2022-2025** — **Senior Postdoctoral Research Associate -** Institute of Environment
Florida International University, *Miami, FL* (Advisors: Dr. Rolando O. Santos, Dr. Jennifer S. Rehage)

**2020-2022** — **Postdoctoral Research Associate** – Institute of Environment
Florida International University, *Miami, FL* (Advisors: Dr. Rolando O. Santos, Dr. Jennifer S. Rehage)

**2016-2020** — **University Doctoral Fellow** – Ecosystem Ecology Lab
University of Louisiana at Lafayette, *Lafayette, Louisiana*

**2019-2020** — **Acoustic Telemetry Technician**
Jesco Environmental, *Portsmouth, New Hampshire*

**2019-2020** — **Menhaden Port Sampler**
Gulf States Marine Fisheries Commission, *Abbeville, Louisiana*

**2016** — **ACES Intern** – Lab for Advanced Sensing
NASA Ames Research Center, *Mountain View, California* (Mentor: Dr. Ved Chirayath)

**2014-2016** — **Graduate Teaching Assistant** – McClintock Lab
University of Alabama at Birmingham, *Birmingham, AL*

**2012-2014** — **Research Assistant** – Earley Animal Behavior Lab
University of Alabama, *Tuscaloosa, AL*

## PUBLICATIONS

59. Herlan JJ, McTague SA, Staniczenko PPA, Loh JM, Santos RO, **James WR**, Gleason ACR, Edwards CB, Gosnell JS, McDonald KC, Friedlander AM, & CF Gaymer (*accepted*) Spatial ecology of coral reefs at Rapa Nui (Easter Island): biotic mechanisms and environmental drivers of coral composition and dispersion along a wave exposure gradient.

58. Costa SV, Jones BLH, **James WR**, Santos RO, & JS Rehage (*accepted*). Guiding practices drive catch-and-release support and habitat concerns in Everglades recreational fishery. *Ocean and Coastal Management*

57. White M, Rodemann JR, **James WR**, Sturges JW, Atkinson CC, Goldner VS, Santos RO, & JS Rehage (*accepted*). Body size and temperature interact with time of day to shape the activity pattern of an ectothermic predator. *Royal Society Open Science*

56. Boucek RE, Kovanda LC, Viadero N, Adams AJ, Rehage JS, Santos RO, & **WR James** (2026) High site fidelity and elevated $\delta^{15}N$ in Bonefish suggest localized wastewater stress in a marine national park. *Estuaries and Coasts* https://doi.org/10.1007/s12237-026-01769-4

55. Linenfelser JO, **James WR**, Rezek RJ, Santos RO, Frankovich TA, Madden C, Nelson JA, & JS Rehage (2026) Isotopic insights into nutrient contributions and transport pathways in north-central Florida Bay. *Estuarine, Coastal and Shelf Science* https://doi.org/10.1016/j.ecss.2026.109765

54. Kabat LJ, Crowl TA, Estruche Santos AS, Cheshire J, **James WR**, & RO Santos (2026) Comparison of tagging techniques for tracking freshwater tropical shrimp: assessment of survival and behavior as a function of tag type. *Crustaceana* https://doi.org/10.1163/15685403-bja10515

53. Rivas N, **James WR**, Bautista V, Sandquist M, Guerriero AF, Arias SB, Mercado A, Rehage JS, & RO Santos (2026) Tipping the balance: Reef community shifts after a regional urchin population collapse. *Coral Reefs* https://doi.org/10.1007/s00338-026-02822-1

52. Sturges JW, **James WR,** Rezek RJ, Santos RO, White M, Badlowski GA, Trabelsi S, Massie, JA, Nelson JA, Trexler JC, & JS Rehage (2026) Seasonality drives green and brown energy pathways in food webs across an ecosystem gradient. *Plos One* https://doi.org/10.1371/journal.pone.0336521

51. Goldner VS, White M, Eggenberger C, Jones AA, Atkinson CC, Rodemann JR, **James WR**, Massie JA, Kroetz AM, O'Donnell PM, Boucek RE, Santos RO, & JS Rehage (2026) Gone but not forgotten: collaborative telemetry network provides insight into out-of-system movements of Common Snook (*Centropomus undecimalis*). *Marine and Coastal Fisheries* https://doi.org/10.1093/mcfafs/vtaf050
**Themed issue: Ecology and Management of Snook Centropomus spp. in the Americas*

50. Massie JA, Rezek RJ, **James WR**, Santos RO, Viadero NM, Boucek RE, & JS Rehage (2025) When the timing is right: linking a seasonal freshwater prey subsidy to the body condition of an estuarine consumer in a subtropical coastal river. *Scientific Reports* https://doi.org/10.1038/s41598-025-25211-0

49. Rodemann JR, White M, **James WR**, Santos RO, Griffin LP, Costa SV, Furman BT, Pittman SJ, Gann D, & JS Rehage (2025) Hierarchical habitat selection by a predatory fish in a patchy seascape. *Scientific Reports* https://doi.org/10.1038/s41598-025-27322-0

48. Geary B, **James WR**, Karubian J, Nelson JA, & PL Leberg (2025) Remote sensing and foraging data illustrate landscape-scale considerations for coastal restoration and avian management. *Ecological Applications* https://doi.org/10.1002/eap.70152

47. White M, **James WR**, Lesser JS, Rezek RJ, Rodemann JR, Boucek RE, Santos RO, & JS Rehage (2025) Linking hydrology, temperature, and energetics: global change implications for the foraging ecology of a generalist predator. *Marine and Coastal Fisheries* https://doi.org/10.1093/mcfafs/vtaf031
**Themed issue: Ecology and Management of Snook Centropomus spp. in the Americas*

46. Massie JA, Rezek RJ, Santos RO, **James WR**, Viadero NM, Boucek RE, & JS Rehage (2025) Long-term patterns in the relative abundance of Common Snook as a factor of shifting environmental

conditions in the Florida Coastal Everglades. *Marine and Coastal Fisheries* https://doi.org/10.1093/mcfafs/vtaf026
**Themed issue: Ecology and Management of Snook Centropomus spp. in the Americas*

45. Viadero NM, Massie JA, Eggenberger CW, **James WR**, Boucek RE, Rezek RJ, RO Santos, & JS Rehage (2025) Between a dry marsh and a salty place: estuarine habitat suitability for a freshwater fish (Florida largemouth bass) and implications for ecosystem restoration and climate change. *Estuaries & Coasts* https://doi.org/10.1007/s12237-025-01561-w

44. Rehage JS, Castillo NA, Distrubell A, Trabelsi S, Santos RO, **James WR,** Rezek R, Cerveny D, Boucek RE, Adams AJ, Fick J, & T Brodin (2025) A multi-estuary assessment of pharmaceutical exposure and risk of pharmacological effects in a recreational fishery. *Environmental Toxicology and Chemistry* https://doi.org/10.1093/etojnl/vgaf125

43. **James WR,** Castillo NA, Distrubell, Trabelsi S, Santos RO, Cerveny D, Rezek RJ, Boucek RE, Adams AJ, Fick J, Brodin T, & JS Rehage (2025) Occurrence of pharmaceuticals in muscle tissue of red drum (*Sciaenops ocellatus*) across subtropical estuaries: comparison to blood plasma and implications for human exposure. *Science of the Total Environment* https://doi.org/10.1016/j.scitotenv.2025.179106

42. Rodemann JR, **James WR**, Rehage JS, Furman BT, Pittman SJ, & RO Santos (2025) Response of submerged aquatic vegetation to a large-scale seagrass die-off: a case study in Florida Bay. *Estuarine, Coastal and Shelf Science* https://doi.org/10.1016/j.ecss.2025.109221

41. Santos RO, **James WR**, Rehage JS, Eggenberger CW, Lesser JS, & CJ Madden (2025) Trophic niche dynamics of two fish mesoconsumers in adjacent coastal habitats with varying nutrient regimes. *Oecologia* https://doi.org/10.1007/s00442-025-05680-w

40. Santos RO, White M, **James WR**, Viadero NM, Massie JA, Boucek RE, & JS Rehage (2025) Cause and consequences of Common Snook (*Centropomus undecimalis*) space use specialization in a subtropical riverscape. *Scientific Reports* https://doi.org/10.1038/s41598-024-82158-4

39. Jones BLH, Santos RO, **James WR**, Shephard S, Adams AJ, Boucek RE, Coals L, Costa SV, Cullen-Unsworth LC, & JS Rehage (2025) Diversity trumps ability: employing the Wisdom of Crowds for fisheries management. *Scientific Reports* https://doi.org/10.1038/s41598-024-84970-4

38. Ziegler SL, **James WR**, Alber M, & JE Byers (2024) Seascape structure likely influences resource use of marsh-associated estuarine consumers over a small spatial scale. *Marine Ecology Progress Series* https://doi.org/10.3354/meps14755

37. Castillo NA, Santos RO, **James WR**, Rezek RJ, Cervany D, Boucek RE, Fick J, Brodin T, & JS Rehage (2024) Widespread pharmaceutical exposure at concentrations of concern for a subtropical coastal fishery: Bonefish (*Albula vulpes*). *Marine Pollution Bulletin* https://doi.org/10.1016/j.marpolbul.2024.117143
(Pre-print SSRN Electronic Journal doi: 10.2139/ssrn.4394268)

36. Kahmann G, Rehage JS, Massie JA, Nelson JA, Santos RO, Viadero NM, **James WR**, Boucek RE, Crane DP, & RJ Rezek (2024) Trophic niche of a nonnative invader and environmental drivers of its increasing populations in the coastal Everglades. *Biological Invasions* https://doi.org/10.1007/s10530-024-03444-w

35. Castillo NA, Santos RO, **James WR**, Rezek RJ, Cerveny D, Boucek RE, Adams AJ, Fick J, Brodin T, & JS Rehage (2024) Differential tissue distribution of pharmaceuticals in a wild subtropical marine fish. *Aquatic Toxicology* https://doi.org/10.1016/j.aquatox.2024.107064

34. Castillo NA, **James WR**, Santos RO, Rezek RJ, Cerveny D, Boucek RE, Adams AJ, Trabelsi S, Distrubell A, Sanquist M, Fick J, Brodin T, & Rehage JS (2024) Identifying pathways of pharmaceutical exposure in a mesoconsumer marine fish. *Journal of Hazardous Materials* https://doi.org/10.1016/j.jhazmat.2024.135382

33. **James WR**, Furman BT, Rodemann JR, Costa SV, Fratto ZW, Nelson JA, Rehage JS, & RO Santos (2024) Widespread habitat loss leads to ecosystem-scale decrease in trophic function. *Global Change Biology* https://doi.org/10.1111/gcb.17263

32. Jones BLH, Santos RO, **James WR**, Costa SV, Adams AJ, Boucek RE, Coals LRM, Cullen-Unsworth LC, Shephard S, & JS Rehage (2024) Lack of evolution in fisheries-associated Indigenous and local knowledge research reveals opportunities for growth. *Fish and Fisheries* https://doi.org/10.1111/faf.12831

31. Raoult V, Phillips AA, Nelson JA, Niella Y, Skinner C, Tilcock MB, Burke PJ, Szpak P, **James WR**, & C Harrod (2024) Why aquatic scientists should use sulfur stable isotope ratios ($\delta^{34}S$) more often. *Chemosphere* https://doi.org/10.1016/j.chemosphere.2024.141816

30. Castillo NA, **James WR**, Santos RO, Cerveny D, Boucek RE, Adams AJ, Goldberg T, Campbell L, Perez AU, Schmitter-Soto JJ, Lewis JP, Fick J, Brodin T, & JS Rehage (2024) Understanding pharmaceutical exposure and the potential for effects in marine biota: a survey of bonefish (*Albula vulpes*) across the Caribbean Basin. *Chemosphere* https://doi.org/10.1016/j.chemosphere.2023.140949

29. Gervasi CL, Karnauskas M, Rios A, Santos RO, **James WR**, Rezek RJ, & JS Rehage (2023) Rapid approach for assessing an unregulated fishery using a series of data-limited tools. *Marine and Coastal Fisheries* https://doi.org/10.1002/mcf2.10270
**2023 Best Paper in Marine and Coastal by American Fisheries Society*

28. Yando ES*, Jones SF*, **James WR***, Colombano DD, Montemayor DI, Nolte S, Raw JL, Ziegler SL, Chen L, Daffonchio D, Fusi M, Rogers K, & L Sergienko. (2023) An integrative salt marsh conceptual framework for global comparisons. *Limnology and Oceanography Letters* https://doi.org/10.1002/lol2.10346
*Contributed equally

27. Iporac LAR, **James WR**, & Collado-Vides L (2023) Characterizing potential resource use of sargasso-dominant sea wrack by terrestrial invertebrate fauna during sargasso influxes in South Florida. *Estuarine, Coastal and Shelf Science* https://doi.org/10.1016/j.ecss.2023.108414

26. Rezek RJ, Massie JA, Nelson JA, Santos RO, Viadero NM, **James WR**, Boucek RE, & JS Rehage (2023) The effects of temperature and flooding duration on the structure and magnitude of a floodplain prey subsidy. *Freshwater Biology* https://doi.org/10.1111/fwb.14145

25. Rodemann JR, **James WR**, Rehage JS, Baktoft H, Costa SV, Ellis RD, Gonzalez L & RO Santos (2023) Residency and fine-scale habitat use of juvenile goliath grouper (*Epinephelus itajara*) in a mangrove nursery. *Bulletin of Marine Science* https://doi.org/10.5343/bms.2022.0061

24. Calhoun-Grosch S, Foster EM, **James WR**, Santos RO, Rehage JS, & JA Nelson (2023) Trophic niche metrics reveal long-term shift in Florida Bay food webs. *Ecosystems* https://doi.org/10.1007/s10021-023-00825-5

23. Massie JA, Santos, RO, Rezek, RJ, **James, WR**, Viadero, NM, Boucek, RE, Blewett DA, Trotter AA, Stevens PW, & JS Rehage (2022). Primed and cued: long-term acoustic telemetry links interannual and seasonal variations in freshwater flows to the spawning migrations of Common Snook in the Florida Everglades. *Movement Ecology* https://doi.org/10.1186/s40462-022-00350-5

22. Gervasi CL, Nelson JA, Swart PK, Santos RO, Rezek RJ, **James WR**, Jefferson AE, Drymon JM, Carroll J, Boucek RE, & JS Rehage (2022) Otolith stable isotope micro-sampling to discriminate poorly studied stocks: Crevalle Jack in the eastern Gulf of Mexico. *Estuarine, Coastal and Shelf Science* https://doi.org/10.1016/j.ecss.2022.108130

21. **James WR,** Bautista V, Rezek RJ, Zink IC, Rehage JS, & RO Santos (2022) A review of the potential impacts of commercial inshore pink shrimp fisheries on recreational flats fisheries in Biscayne Bay, FL, USA. *Environmental Biology of Fishes* https://doi.org/10.1007/s10641-022-01319-4
**Special issue: Conservation Connections: Science informing management for flats fisheries*

20. Rezek RJ, **James WR**, Bautista V, Zink IC, Rehage JS, & RO Santos (2022) Temporal trends of Biscayne Bay pink shrimp fisheries catch and economic indicators. *Environmental Biology of Fishes* https://doi.org/10.1007/s10641-022-01314-9
**Special issue: Conservation Connections: Science informing management for flats fisheries*

19. **James WR,** Santos RO, Rodemann JR, Rezek RJ, Fratto ZW, Furman BT, Hall MO, Kelble CR, Rehage JS, & JA Nelson (2022) Widespread seagrass die-off has no legacy effect on basal resource use of seagrass food webs in Florida Bay, USA. *ICES Journal of Marine Science fsac112,* https://doi.org/10.1093/icesjms/fsac112

18. Santos RO, **James WR**, Nelson JA, Rehage JS, Serafy J, Pittman SJ, & D Lirman (2022) Influence of seascape spatial pattern on the trophic niche of an omnivorous fish. *Ecosphere* https://doi.org/10.1002/ecs2.3944

17. **James WR**, Santos RO, Rehage JS, Doerr JC, & JA Nelson (2021) *E*-scape: consumer specific landscapes of energetic resources derived from stable isotope analysis and remote sensing. *Journal of Animal Ecology* https://doi.org/10.1111/1365-2656.13637
(Pre-print bioXriv 2020 doi: https://doi.org/10.1101/2020.08.03.234781)

16. Rodemann JR, **James WR**, Santos RO, Furman BT, Fratto ZW, Bautista V, Hernandez JL, Viadero NM, Linenfelser JO, Lacy LA, Hall MO, Kelble CR, Kavanaugh C, & JS Rehage (2021) Impact of extreme disturbances on suspended sediment in Western Florida Bay: Implications for seagrass resilience. *Frontiers in Marine Science* https://doi.org/10.3389/fmars.2021.633240
**Special issue: Environmental Threats to the State of Florida—Climate Change and Beyond*

15. Gervasi CL, Santos RO, Rezek RJ, **James WR**, Boucek RE, Bradshaw C, Kavanagh C, Osborne J, & JS Rehage (2021) Bottom-up conservation: Using translational ecology to inform conservation priorities for a recreational fishery. *Canadian Journal of Fisheries and Aquatic Sciences* https://doi.org/10.1139/cjfas-2021-0024

14. Kimball ME, Connolly RM, Alford SB, Colombano DD, **James WR**, Kenworth MD, Norris GS, Ollerhead J, Ramsden S, Rehage JS, Sparks EL, Waltham NJ, Worthington TA, & MD Taylor (2021) Novel applications of technology for advancing tidal marsh ecology. *Estuaries & Coasts* doi: 10.1007/s12237-021-00939-w
**Special issue: Concepts and controversies in tidal marsh ecology revisited.*

13. Pittman SJ, Yates KL, Bouchet PJ,… **James WR**… (34 additional authors) (2021) Seascape ecology: Identifying research priorities for an emerging ocean sustainability science. *Marine Ecology Progress Series* https://doi.org/10.3354/meps13661

12. Nelson JA, Harris JM, Lesser JS, **James WR**, Suir SM, & WP Broussard III (2020) New mapping metrics to test functional response of food webs to coastal restoration. *Food webs* https://doi.org/10.1016/j.fooweb.2020.e00179
**Special issue: Restoration Initiatives Viewed Through a Lens of Food Web Structure and Dynamics*

11. Jones SF, Stagg CL, Yando ES, **James WR**, Buffington KJ, & MW Hester (2020) Stress gradients interact with disturbance to reveal alternative states in salt marsh: Multivariate resilience at the landscape scale. *Journal of Ecology* https://doi.org/10.1111/1365-2745.13552
**Special issue: Reconciling resilience across ecological systems, species and subdisciplines.*

10. Baker R, Taylor MD, Beck MW, Cebrian J, Colombano DD, Connolly RM, Currin C, Deegan LA, Feller IC, Gilby BL, Kimball ME, Minello TJ, Rozas LP, Simenstad C, Turner RE, Waltham NJ, Weinstein MP, Ziegler SL, zu Ermgassen PSE, … **James WR**… (29 additional authors) (2020) Fisheries rely on threatened salt marshes. *Science* doi: 10.1126/science.abe9332

9. **James WR**, Topor ZM, & RO Santos (2020) Seascape structure influences the community structure of marsh nekton. *Estuaries & Coasts* doi: 10.1007/s12237-020-00853-7
**Special issue: Concepts and controversies in tidal marsh ecology revisited.*

8. Harris JM[†], **James WR**[†], Lesser JS[†], Doerr JC, & JA Nelson (2020) Foundation species shift alters the energetic landscape of marsh nekton. *Estuaries & Coasts* doi: 10.1007/s12237-020-00852-8
[†]Contributed equally
**Special issue: Concepts and controversies in tidal marsh ecology revisited.*

7. Lesser JS, **James WR**, Stallings CD, Wilson RM, & JA Nelson (2020) Trophic niche size and overlap decrease with increasing ecosystem productivity. *Oikos* doi: 10.1111/oik.07026

6. **James WR**, JS Lesser, SY Litvin, & JA Nelson (2020) Assessment of food web recovery following restoration using resource use metrics. *Science of the Total Environment* doi: 10.1016/j.scitotenv.2019.134801.

5. Nelson JA, Lesser JS, **James WR**, Behringer DP, Furka V, & JC Doerr (2019) Food web response to foundation species change in a coastal ecosystem. *Food Webs* e00125 https://doi.org/10.1016/j.fooweb.2019.e00125

4. Eggenberger CW, Santos RO, Frankovich T, **James WR,** Madden C, Nelson JA, & JS Rehage (2019) Coupling telemetry and stable isotope techniques to unravel movement: Snook habitat use across variable nutrient environments. *Fisheries Research* https://doi.org/10.1016/j.fishres.2019.04.008

3. **James WR,** Styga JM, White S, Marson KM, & RL Earley (2018) Developmentally plastic responses to predation threat in the mangrove rivulus fish (*Kryptolebias marmoratus*): behavior and morphology. *Evolutionary Ecology* https://doi.org/10.1007/s10682-018-9952-5

2. **James WR** & JB McClintock (2017) Anti-predator responses of amphipods are more effective in the presence of conspecific chemical cues. *Hydrobiologia* https://doi.org/10.1007/s10750-017-3191-6

1. Smith KE, Aronson RB, Steffel BV, Amsler MO, Thatje S, Singh H, Anderson J, Brother CJ, Brown A, Ellis DS, Havenhand JN, **James WR**, Mosknes P, Randolph AW, Sayre-McCord T, & JB McClintock (2017) Climate change and the threat of novel marine predators in Antarctica. *Ecosphere* https://doi.org/10.1002/ecs2.2017

***Manuscripts under review***

**James WR**, White M, Rodemann JR, Badlowski GA, Sturges JW, Leavitt H, Lesser JS, Bautista V, Castillo NA, Costa SV. Distrubell A, Eggenberger CW, Kabat LJ, Linenfelser JO, Rivas N, Trabelsi S, Rehage JS, & RO Santos (*in review*) Seasonality in species trophic niche width and position alters community niche occupancy.

Rodemann JR, **James WR**, Rehage JS, Santos RO, Fratto ZW, Strazisar T, Madden CJ, & TA Frankovich (*in revision*) Drivers of spatiotemporal SAV dynamics in the coastal lakes of Everglades National Park.

White M, **James WR**, Lemoine NP, Allgeier JE, Stier AC, Emery KA, Castorani MCN, Capone D, Stajner A, Enright L, Grier S, Cawley G, Spivak A, Nelson JA, Hopcroft RR, Chen A, Lyon NJ, Kui L, Caselle JE, Strickland BA, Peters J, Rehage JS, & D Burkepile (*in revision*). Scale-dependent effects of species richness and asynchrony regulate the temporal stability of consumer-mediated nutrient dynamics.

Herrera MA, **James WR**, Boucek RE, Andres MJ, Ault JER, Bassos-Hull K, Brownscombe J, Dance M, Ellis RD, Fox D, Gardiner J, Griffin LP, Kroetz AM, Lamont MM, Locascio J, Lowerre-Barbieri SK, Morley D, Poulakis GR, Powers SP, Ramsden S, Rooker JR, Soldevilla M, Stephens B, Wells RJD, Wilkinson KA, Wooley AK, Santos RO, Drymon JM, Rehage JS, & Gervasi CL (*in revision*) From Florida to Texas: Seasonal movements of Crevalle Jacks informed by collaborative acoustic telemetry.

Rivas N, **James WR**, Bautista V, Mercado-Molina AE, & RO Santos (*in review*) Coral species demography reshaped after keystone herbivore loss.

Badlowski GA, **James WR**, Nation C, Coppola M, Rehage JS, & RO Santos (*in review*). Seascape-scale habitat amount and mangrove connectivity, not patch composition, structure seagrass prey communities.

Doomstorm D, Gomez-Gonzalez E, Acosta Rodríguez VN, Figueroa A, **James WR**, Rehage J, Santos R, Ziegler SL, Bowen JL, Johnson DS, Spivak AC, & JA Nelson (*in revision*). Expanding globally, engineering locally: Latitudinal consistency and localized ecological effects of fiddler crab bioturbation.

**FUNDING ($9,293,027.93 total funding)**

**2026** ReWater: Biscayne Bay Water Quality Feasibility Study – CoPI: Recommended for funding by US Environmental Protection Agency South Florida Geographic Program (2026-2029), $999,744

Salinity Thresholds and Submerged Vegetation Resilience: Preventing Lake-to-Bay Bloom Cascades in Florida Bay – CoPI: Recommended for funding by National Parks Service, $1,055,582 (2026-2031)

**2025** A holistic assessment of the Florida Keys flats food web – PI: Funded by Bonefish and Tarpon Trust, $277,000 (2025-2027)

Coastal Riverine Fish Dynamics and Marsh-Mangrove Habitat Use in Shark River Slough, Florida – Co-PI: Funded by United States Army Core of Engineers REstoration COordination and VERification (RECOVER) program of the Comprehensive Everglades Restoration Plan (CERP), $1,006,142 (2025-2030)

**2024** Understanding seascapes, susceptibility, and food web impacts of the fish spinning phenomenon in the Florida Keys – PI: Funded by Bonefish and Tarpon Trust, $227,053 (2024-2025)

SAV Mapping of Biscayne Bay, Florida for SIMM – Co-PI: Funded by Florida Fish and Wildlife Commission, $48,000 (2024-2025)

The role of coastal lakes in Florida Bay as fish habitat for critical recreational fisheries: Movement and trophic research – Co-PI: Funded by South Florida Water Management District, $98,847 (2024-2025)

Assess Impacts of Trawling on Species and Habitats within Biscayne Bay – Co-PI: Funded by National Park Service, $245,804 (2024-2026)

Assessing spatiotemporal responses to CERP by coastal fish assemblages across Everglades National Park – Co-PI: Funded by National Park Service, $225,101 (2024-2026)

Assessing the resilience of coastal fisheries to press and pulse disturbances in Everglades National Park through a comprehensive analysis of fisheries-dependent records – Co-PI: Funded by National Park Service, $283,498 (2024-2026)

Enhancing our understanding of seagrass health: critical thresholds and effects of land based pollution from nutrients & emergent contaminants in Biscayne Bay – Co-PI: Funded by US Environmental Protection Agency South Florida Geographic Program, $649,956 (2024-2025)

**2023** A nationwide survey of pharmaceutical exposure in national parks – PI: Funded by National Park Service, $60,000 (2023-2025)

Spatiotemporal variation of Permit diet and resource use across the Florida Keys – PI: Funded by Bonefish and Tarpon Trust, $5,000 (2023-2024)

Developing a standardized assessment protocol for flats fisheries – Co-PI: Funded by Bonefish and Tarpon Trust, $216,406 (2023-2024)

Role of coastal lakes as habitat for recreational fisheries – Co-PI: Funded by South Florida Water Management District (2023) $142,482

Increasing knowledge of Biscayne Bay's science status and trends: Identify monitoring course of action through a synthesis of experts' knowledge – Co-PI: Funded by Florida Sea Grant (2023), $9,708

**2022** Seagrass seascape state, stability, and function in relation to water quality in Biscayne Bay – PI: Funded by US Environmental Protection Agency South Florida Geographic Program (2022-24), $347,738

RAPID: Here we go again - The fate of *Diadema antillarum* and Caribbean reefs in the 21st century – Co-PI: Funded by National Science Foundation, Biological Oceanography (2022-2023), $199,676

An examination of the risk of pharmaceuticals in South Florida: extent and pathways of exposure in a valuable recreational fishery, bonefish – Co-PI: Funded by US Environmental Protection Agency South Florida Geographic Program (2022-24), $292,530

A Statewide survey of pharmaceutical exposure: Red Drum in 9 Florida estuaries – Co-PI: Funded by Bonefish and Tarpon Trust (2022), $158,149

Quantifying ecological importance, historical threat exposure and resilience potential for coral reef Areas of Special Interest in Puerto Rico's Northeast Marine Corridor to support efficient conservation goals – Co-PI: Funded by Puerto Rico Sea Grant College Program (2022-24), $181,942

Redesign of the Everglades fishing guide reporting system – Co-PI: Funded by Everglades National Park (2022-2023), $109,028

Role of coastal lakes as habitat for recreational fisheries – Co-PI: Funded by South Florida Water Management District (2022), $170,000

An Examination of Wastewater Treatment in Miami-Dade and Monroe Counties: a focus on the removal of pharmaceuticals and potential threat to water quality in the Everglades and Biscayne Bay – Co-PI and REU mentor: Funded by National Science Foundation, Florida Coastal Everglades Long-term Ecological Research (DEB-2025954), Research Experiences for Undergraduates (REU) (2022), $6,750

**2021** FISHSCAPE Project (Fish In Seagrass Habitats: Seascape Connectivity Across Protected Ecosystems)– Postdoc and grant writer: Funded by NOAA National Centers for Coastal Ocean Science (2021-25), $1,916,914

Biscayne Bay shrimp commercial harvest and its potential ecological impacts on the recreationally fish species – Co-PI: Funded by Bonefish and Tarpon Trust (2021), $16,575

The effects of the legacies of a large-scale seagrass die-off on energy resources and foraging of a top consumer – Co-PI and REU mentor: Funded by National Science Foundation, Florida Coastal Everglades Long-term Ecological Research (DEB-2025954), Research Experiences for Undergraduates (REU) (2021), $6,750

**2016-2020** UL University Fellow; $90,000
UL Graduate Student Organization Research Supply Grant; $1060 total
**2017-2019** UL Graduate Student Organization Travel Grant; $1240 total
**2017** Coastal and Estuarine Research Federation Travel Grant; $300
**2016** Ted Beaulieu Sr. Coastal Conservation Association Louisiana Scholarship; $5000
**2015** UAB Biology Department Travel Grant; $400
Society for Integrative and Comparative Biology Grant in Aid of Research; $850
Sigma Xi Grant in Aid of Research; $400

***Proposals pending or not selected for funding***

**2026** ReWater: A Demonstration Project for Improving Water Quality to Support Biscayne Bay Restoration – Co-PI: Submitted to US Environmental Protection Agency South Florida Geographic Program, $999,744 (*Pending*)

**2024** An evaluation of climate-driven changes in South Atlantic Fisheries – PI: Submitted to South Atlantic Fisheries Management Council, $324,874 (*Not selected for funding*)

**2023** Accentuating natural defenses: strategic mangrove planting to enhance coastal wetland resilience – Co-PI: Submitted to NOAA Transformational Habitat Restoration, $3,084,004 (*Not selected for funding*)

Assessing the functionality and coral demographic performance of reefs as a function of environmental and sediment-turf conditions to inform restoration suitability across South Florida – Co-PI: Submitted to US Environmental Protection Agency South Florida Geographic Program, $644,158 (*Not selected for funding*)

Influence of habitat degradation on the ecology, behavior and health status of coastal bottlenose dolphins in Biscayne Bay - Co-PI: Submitted to US Environmental Protection Agency South Florida Geographic Program, $391,573 (*Not selected for funding*)

Quantifying coral demographic performance across trait space: Assessing functionality of coral reefs across South Florida Reef Tract – Co-PI: Submitted to Florida Sea Grant College Program, $185,502 (*Not selected for funding*)

**2022** Visual identification, population assessment, and habitat suitability of juvenile Goliath Grouper (*Epinephelus itajara*) in South Florida – Co-PI: Submitted to NOAA Marine Fisheries Initiative, $524,566 (*Not selected for funding*)

Examining juvenile Goliath Grouper habitat use and abundance patterns in Florida Bay and the Florida Keys – Co-PI: Submitted to NOAA Cooperative Research Program, $198,623 (*Not selected for funding*)

Standardizing and comparing food web structure and function across LTER sites - PI: Submitted to National Center for Ecological Analysis and Synthesis, $109,200 (*Not selected for funding*)

**2021** The influence of nearshore inputs (freshwater inflows) on the habitat use of an ecosystem indicator species: Spotted Seatrout in Florida Bay – Co-PI: Submitted to the US Coastal Research Program, $279,158 (*Not selected for funding*)

Assessing the resilience of coastal fisheries through a comprehensive analysis of fisheries-dependent records: the Everglades as a case study – Co-PI: Submitted to the US Coastal Research Program, $283,844 (*Not selected for funding*)

Quantifying coral demographic performance within a functional trait space to inform coastal ecosystem conservation – Co-PI: Submitted to the US Coastal Research Program, $299,005 (*Not selected for funding*)

**2020** Critical data for Dolphin (*Coryphaena hippurus*) fisheries sustainability: Assessing stock structure, properties, trends & vulnerabilities across U.S. jurisdictions – Co-PI: Submitted to NOAA Marine Fisheries Initiative, $524,000 (*Not selected for funding*)

## MENTORING

**2022-Present** **Graduate Mentorship and Committee Service**
*Co-Major Advisor – Alia Jones, PhD Earth & Environment, FIU, 2024-present*
*Co-Major Advisor – Camille Demaire, PhD Earth & Environment, FIU, 2024-present*
*Committee Member – McKenzie Zapata, PhD Earth & Environment, FIU, 2024-present*
*Committee Member – Hannah-Marie Lamle, PhD Biology, FIU, 2023-present*
*Committee Member – Gina Badlowski, PhD Biology, FIU, 2022-present*
*Committee Member – Marianna Coppola, PhD Biology, FIU, 2022-present*
*Committee Member – Sophia Costa, PhD Biology, FIU, 2022-present*
*Committee Member – Andy Distrubell, MS Earth & Environment, FIU, 2023-2025*
*Committee Member – Valentina Bautista, MS Biology, FIU, 2022-2024*

**2020-Present** **Graduate and Undergraduate Student Research Mentor,** Florida International University – Part of the lab leadership team that has mentored 21 graduate students, 9 REUs, and 12 undergraduate research technicians

**2016-2020** **Undergraduate Student Research Mentor,** University of Louisiana at Lafayette
Laura McDonald – *The energy storage of nekton species in coastal Louisiana habitats*
- Presented at the Coastal and Estuarine Research Federation meeting (CERF 2019) in Mobile, AL
- Presented research at the UL Biology undergraduate research conference

*The effect of feeding mode on energy storage in three fish species*
- Presented research at the UL honors program undergraduate research conference

Blake Kitchen - *The role of food quality in growth and energy storage in killifish*

**2015-2016** **Undergraduate Student Research Mentor,** University of Alabama at Birmingham
Ciara Duncan - *Predators effect the foraging behavior of the amphipod Hyalella azteca*
- Received 2nd place award for poster presentation at UAB Undergraduate Student Research Days

Keegan McFarland and Deontae Mitchell - *Transgenerational effects of predator exposure of the amphipod Hyalella azteca*

## TEACHING EXPERIENCE

***Teaching***

**2022-Present** *R workshop on population and community ecological modeling* – Co-Instructor, Florida International University

**2025** *Using a multivariate tool to assess population and community dynamics* – Instructor; Florida International University

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| **2024** | *Hypervolume modelling: a multivariate tool for seagrass ecosystem assessments* – Workshop Convener, 15th International Seagrass Biology Workshop, Naples, Italy |
| **2020** | *Fish Ecology and Management* – Teaching Assistant, University of Louisiana at Lafayette |
| **2019** | *Introduction to R* – Teaching Assistant, University of Louisiana at Lafayette |
| **2018** | *Fundamentals of Biology II Lab* – Teaching Assistant, University of Louisiana at Lafayette |
| **2017** | *Landscape Ecology* – Teaching Assistant, University of Louisiana at Lafayette |
| **2014-2016** | *Introductory Biology Lab* – Teaching Assistant, University of Alabama at Birmingham |

***Guest Lectures***

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| **2021** | "Intro into R: data types, loops, functions, random sampling, and plotting" (2 part), *Advanced Ecology: Populations and Communities* |
| | "Habitat loss and fragmentation effects in seascapes", *Biodiversity Across Ecosystems* |
| **2020** | "Introduction to food webs", *Intro to Coastal Ecology* |
| **2019** | "Data wrangling in R" (2 part), *Quantitative Ecology* |
| | "Effect of landscape pattern on ecological processes" (3 part), *Ecosystem landscape ecology* |
| | "Maximum entropy ecological niche modeling in R", *Ecological models and data* |
| **2015** | "Long distance prey finding", *Chemical Ecology* |

**SELECTED PRESENTATIONS** (**undergraduate mentee,* graduate student mentee)

**2026**

**James WR**, Badlowski GA, Coppola M, Malone MA, Harborne AR, & RO Santos. A system level assessment of trophic function reveals importance of shallow seagrass beds for coral reef fish; International Coral Reef Symposium; Auckland, New Zealand

**James WR**, Coppola M, Badlowski GA, Rehage JS, & RO Santos. Spatiotemporal patterns of seagrass seascape state and stability in Biscayne Bay; Biscayne Bay Health Summit; Miami, FL

**James WR**, Coppola M, Badlowski GA, Rehage JS, & RO Santos. Spatiotemporal patterns of seagrass seascape state and stability across multiple scales in Biscayne Bay; Benthic Ecology Meeting; Virginia Beach, VA

**James WR**, Balowski GA, Coppola M, Malone MA, Harborne AR, & RO Santos; Trophic connectivity across reefscapes: The importance of shallow seagrass beds for coral reef fish; Benthic Ecology Meeting; Virginia Beach, VA

Badlowski GA, Santos RO, & **WR James**. Influence of water quality of trophic function of seagrass ecosystems; Benthic Ecology Meeting; Virginia Beach, VA

Jones AA, Rehage JS, Santos RO, Rodemann JR, Lesser JS, LaBadie J, Boucek RE, & **WR James**. Regional variations in trophic niche and resource use provide context for decline in recreational fishery; Benthic Ecology Meeting; Virginia Beach, VA

**2025**

**James WR**, Badlowski GA, Coppola M, Rodemann JR, Lesser JS, Rehage JS, & RO Santos. A system level assessment of spatiotemporal changes of trophic function using *E*-scapes across seagrass seascapes in Biscayne Bay; Coastal and Estuarine Research Federation Biennial Conference; Richmond, VA

Jones AA, Rehage JS, Santos RO, Rodemann JR, Lesser JS, LaBadie J, Boucek RE, & **WR James**. Regional shifts in diet and resource use as potential cause of declines in quality of key seagrass flat fishery; Coastal and Estuarine Research Federation Biennial Conference; Richmond, VA

**James WR**, Jones AA, Rehage JS, Santos RO, Rodemann JR, Lesser JS, & RE Boucek. Update of the flats food web study; Bonefish and Tarpon Trust Science Symposium; Ft. Lauderdale, FL

**James WR**, Jones AA, Rehage JS, Santos RO, Rodemann JR, Lesser JS, & RE Boucek. Seascape and food web impacts of the fish spinning phenomenon in the Florida Keys; Florida Keys Marine Science Conference; Coral Springs, FL

Jones AA, Rehage JS, Santos RO, Rodemann JR, Lesser JS, LaBadie J, Boucek RE, & **WR James**. Regional differences in permit diet and resource use suggest shifts in seagrass flat food webs; Florida Keys Marine Science Conference; Coral Springs, FL

**James WR,** Rehage JS, & RO Santos. The impact of mass mortality of keystone herbivores on food web energy flow and community trophic niche dynamics; Ecological Society of America Annual Meeting; Baltimore, MD

**James WR,** An ecosystem approach to fisheries in a changing world. University of Hawai'i at Mānoa (Invited)

**2024** **James WR,** Santos RO, Hensel E, Baustista V, Kabat L, White M, Strickland B, Kominoski J, & CJ Patrick. Coastal fish communities' trait composition shifts in response to tropical cyclones, Hurricane Ecosystem Response Synthesis Seminar Series (Invited)

**James WR**, Furman BT, Rodemann JR, Costa SV, Fratto ZW, Nelson JA, Rehage JS, & RO Santos. Widespread seagrass loss leads to ecosystem-scale loss of trophic function, World Seagrass Conference and International Seagrass Biology Workshop; Naples, Italy

**James WR,** An ecosystem approach to fisheries in a changing world. Chesapeake Biological Laboratory, University of Maryland Center for Environmental Science (Invited)

**James WR**, Castillo NA, Distrubell A, Trabelsi S, Santos RO, & JS Rehage. A statewide survey of pharmaceutical exposure in a valuable recreational fishery, Red Drum (*Sciaenops ocellatus*), across 9 Florida Estuaries; Gulf of Mexico Conference; Tampa, FL

**James WR**, Incorporating ecosystem function into fisheries management. University of Vermont (Invited)

**2023** **James WR**, Coppola M, Badlowski G, Rezek RJ, Rehage JS, & RO Santos. Spatiotemporal patterns of seagrass seascape state and stability in South Florida. Coastal and Estuarine Research Federation Biennial Conference; Portland, OR

**James WR,** Rodemann JR, Badlowski G, Bautista V, Castillo NA, Costa SV, Distrubell A, Eggenberger CW, Kabat L, Linenfelser JO, Rivas N, Sandquist M, Sturges J, Trabelsi S, White M, Rehage JS, & RO Santos. Seasonal variation in trophic niche size and overlap in a seagrass food web. Benthic Ecology Meeting; Miami, FL

**2022** **James WR,** Drivers of ecosystem function: from pharmaceuticals to seascapes. University of Memphis (Invited)

**James WR,** *Bautista V**, Rezek RJ, Zink IC, Rehage JS, & Santos RO. The Biscayne Bay commercial shrimp harvest and its potential ecological impacts of recreational fish species. Bonefish & Tarpon Trust Science Symposium and Flats Expo; Palm Beach, FL

**James WR**, Rodemann JR, Fratto ZW, Furman BT, Rehage JS, & Santos RO. Using *E*-scapes to quantify the change in trophic function of seagrass habitats in response to large-scale seagrass die-off. International Seagrass Biology Workshop; Annapolis, MD

**James WR,** Rodemann JR, *Gonzalez L**, Baktoft H, Ellis RD, Santos RO, & Rehage JS. Fine scale movement of juvenile Goliath Grouper. FACT Network; Melbourne, FL

**2021** **James WR,** A seascape approach to consumer movement, distribution, and resource use. Virtual Colloquium, Department of Biological Sciences Walla Walla University (Invited)

**James WR,** Rodemann JR, Rezek RJ, Fratto ZW, Furman BT, Rehage JS, & Santos RO. Using *E*-scapes to quantify the change in energetic resource distribution and foraging of seagrass consumers. Coastal and Estuarine Research Federation Biennial Conference

*Ajavon A**, **James WR**, Rodemann JR, Rezek RJ, Fratto ZW, Furman BT, Rehage JS, & Santos RO. The role of seascape configuration on seagrass food web structure and function. Coastal and Estuarine Research Federation Biennial Conference

**James WR**, Rehage JS, & Santos RO. Mapping energy flow: *E*-scapes as a system-level tool to evaluate restoration and ecosystem function. Greater Everglades Ecosystem Restoration

**2019** **James WR**, Geary B, Karubian J, Leberg PL, & Nelson JA. Tracking foraging behavior using consumer-specific energetic landscapes. Coastal and Estuarine Research Federation Biennial Conference; Mobile, AL

*McDonald L*,* **James WR**, Robinson KL, & Nelson JA. The energy storage of nekton species across a continuum of coastal Louisiana habitats. Coastal and Estuarine Research Federation Biennial Conference; Mobile, AL

Nelson JA, Lesser JS, & **James WR**. A new niche for isotope ecology: Combing MixSIAR and hypervolume metrics to understand resource use. Coastal and Estuarine Research Federation Biennial Conference; Mobile, AL

**James WR**, Santos RO, Rodemann JR, Rehage JS, & JA Nelson. Consumer-specific energetic landscapes in Florida Bay: linking seagrass die off to economically valuable fisheries. Greater Everglades Ecosystem Restoration; Coral Springs, FL

**2018** **James WR**, Lesser JS, Litvin SY & Nelson JA. Assessment of food web recovery following restoration using hypervolume analysis. Gulf Estuarine Research Society Biennial Meeting; Galveston, TX

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| | Lesser JS, **James WR**, Doerr J & Nelson JA. Food web effects of mangrove encroachment on estuarine consumers. Gulf Estuarine Research Society Biennial Meeting; Galveston, TX |
| | **James WR**, Santos RO, Rehage JS, Doerr J & Nelson JA. Consumer specific energetic landscapes derived from stable isotope analysis and habitat cover. LTER Network All Scientists' Meeting; Pacific Grove, CA |
| | **James WR,** Litvin SY & Nelson JA. Measuring food web response to restoration in coastal ecosystems. State of the Coast; New Orleans, LA |
| **2017** | **James WR** & Nelson JA. Food Web Response to Habitat Restoration in Various Coastal Wetland Ecosystems. American Geophysical Union Annual Meeting; New Orleans, LA |
| | **James WR**, Deegan L, Garritt R & Nelson JA. Residence time of water in estuary alters resource use on multiple trophic levels. Coastal and Estuarine Research Federation Biennial Conference; Providence, RI |
| **2016** | Adelson J, **James WR**, Chirayath V & Fringer O. A novel approach for estimating vertical profiles of suspended sediment concentration. Ocean Optics XXIII; Victoria, BC |

## AWARDS AND HONORS

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| **2024** | Best paper in Marine and Coastal Fisheries – American Fisheries Society |
| **2023** | Florida Coastal Everglades LTER Collaborator of the Year |
| **2020** | University of Louisiana Graduate School Research Showcase Paper Competition Winner<br>University of Louisiana Graduate School 'Say it in 6' Winner |
| **2018** | Louisiana Sea Grant Discovery, Integration, Application (LaDIA) Graduate Research Scholar |
| **2017** | Louisiana Sea Grant Coastal Connections 3 Minute Thesis Competition Winner |
| **2016** | National Science Foundation Antarctic Service Medal |

## PROFESSIONAL AND COMMUNITY SERVICE

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| **2022-Present** | **Committee member** – Florida Coastal Everglades LTER Internal Executive Committee |
| **2022-Present** | **Chair –** Florida Coastal Everglades LTER Broadening Participation Committee |
| **2022-Present** | **Committee member** – Florida Coastal Everglades LTER representative for the LTER network Broadening Participation Committee |
| **2015-Present** | **Peer reviewer for:** *Estuarine, Coastal and Shelf Science*, *Scientific Reports*, *Marine Biology*, *Ecological Engineering*, *Proceedings of the Royal Society B*, *Biology Letters*, *Marine Ecology Progress Series*, *Thalassas*, *Invertebrate Biology, Aquatic Biology, Journal of Marine Science and Engineering, Marine Biology Research, Estuaries and Coasts, Wetlands, Applications in Plant Science, Diversity, Plos One, Water, Frontiers in Marine Science, Journal of Applied Ecology, Journal of Experimental Marine Biology and Ecology, Coral Reefs, Ecosphere, Freshwater Biology, Science of the Total Environment, Environmental Pollution, Ecological Informatics, Food Webs, Proceedings of the National Academy of Sciences, Journal of Biogeography* |
| **2026** | **Expert committee member**, Canadian Foundation for Innovation John R. Evans Leaders Fund |
| **2021-2023** | **Peer reviewer,** British Ecological Society Review College |
| **2022** | **Judge,** Florida International University Biology Graduate Student Symposium |
| **2021** | **Guest Reviewer,** Biodiversity Across Ecosystems Collaborative Online International Learning Student Research Proposals |

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| | **Judge,** Florida International University Biology Graduate Student Symposium |
| **2018-2019** | **Off-campus representative,** Florida Coastal Everglades LTER Graduate Student Executive Board. |
| **2017-2019** | **Organizing Committee,** UL Lafayette Graduate Student Symposium |
| **2018** | **Workshop Organizer,** "Incorporating the spatial ecology of consumers into long-term research" at LTER Network All Scientists' Meeting |
| | **Reviewer,** Ecological Society of America Student Section Travel Award |
| **2015** | **Science Fair Judge,** Avondale Elementary School Science Fair |
| | **Student mentor,** Avondale Elementary School Science Fair |
| | **Participant,** UAB in Antarctica |

## PROFESSIONAL DEVELOPMENT

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| **2024** | **Functional Trait Analysis in R**, Physalia |
| **2023** | **Fieldwork Harassment Prevention and Intervention**, FieldFutures |
| | **Identifying and Eliminating Micro-aggressions**, Office to Advance Women, Equity & Diversity, FIU |
| | **Safe Zone Advocate Training**, Office to Advance Women, Equity & Diversity, FIU |
| **2022** | **STRIDE: Strategies and Tactics for Recruitment to Increase Diversity and Excellence**, Office to Advance Women, Equity & Diversity, FIU |
| | **Zero-inflated GLM and GLMM in R,** Highland Statistics Ltd. |
| | **Movement Ecology in R**, Physalia Courses |
| **2021** | **Advanced Ecological Niche Modeling Using R**, PR Statistics |
| **2018** | **Louisiana Sea Grant Coastal Immersion Program**, Louisiana Sea Grant |
| **2017** | **Small Boat Operations for Marine Scientists**, Louisiana Universities Marine Consortium (LUMCON) |
| | **Use of Stable Isotopes & Fatty Acids in Aquatic Ecology: Theory & Practice**, Universidad de Antofagasta |
| **2016** | **Advanced Computing for Earth Sciences**, University of Virginia |

## AFFILIATIONS

*Florida Coastal Everglades Long Term Ecological Research Program* –Collaborator
*FIU Institute of Environment CREST Center for Aquatic Chemistry and Environment* – Collaborator
*Hurricane Ecosystem Response Synthesis Network* – Collaborator
*Department of Biology, Florida International University* – Graduate Faculty

## SKILLS

**Field:** Small boat operation; Acoustic telemetry- tagging, deploying, data download; Specimen collection- Seine, gillnet, longline, crab pots, trawl, drop samplers, hook and line

**Computer Software:** R, ArcGIS, QGIS, Python, MATLAB, ImageJ, Jwatcher